

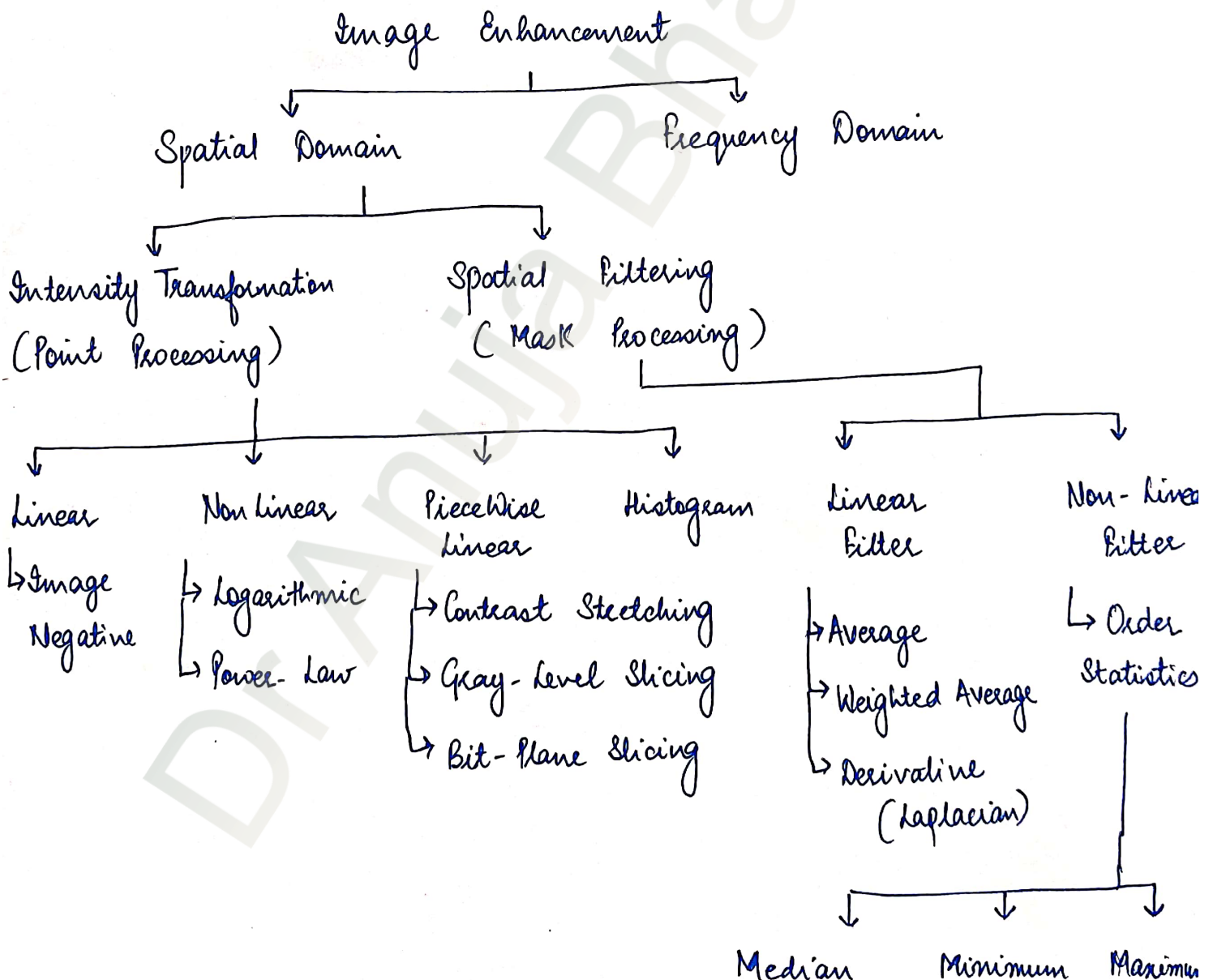
Introduction to Image Enhancement

- Enhancement is process of manipulating an image so that the result is more suitable for a specific application.

Eg: A method is useful for enhancing X-Ray images may not be the best approach for enhancing infrared images.

- Images may be suffer from the following degradations

- Poor contrast due to poor illumination
- Atmospheric disturbances
- Aliasing effect



• There are two broad categories of Image Enhancement approaches (14)

→ Spatial domain - operates directly on the pixels of the image.
- refers to the image plane itself.

→ Frequency domain - operates on the Fourier transform of an image

Basics of Intensity Transformation & Spatial Filtering

• Two principal category of spatial processing are

(i) Intensity Transformation -

Operates on single pixels of an image, principally for the purpose of contrast manipulation & image thresholding.

(ii) Spatial Filtering -

deals with performing operations, such as image sharpening, by working in neighborhood of every pixel in an image.

• Generally, spatial domain techniques are -

→ Computationally more efficient &

→ Require less processing resources to implement.

• The spatial domain process can be denoted by:

$$g(x, y) = T[f(x, y)]$$

where $f(x, y)$ is an input image

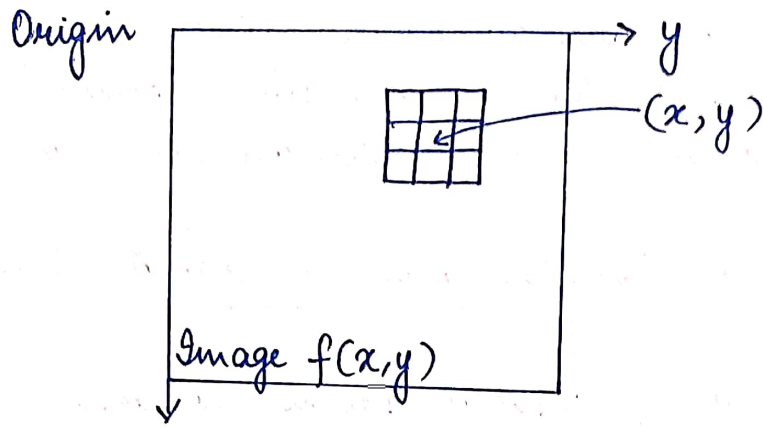
$g(x, y)$ is an output image

T is an operator on f defined over neighborhood of (x, y)

• The operator can be applied to the pixels of a single image or to the pixels of set of images.



- fig shows the basic implementation of above eqn on single image.



- The point (x,y) shown in an arbitrary location in the image, and the small region shown containing the point is a neighborhood of (x,y) .
- The neighborhood is rectangular, centered on (x,y) and much smaller in size than the image.
- The process involves moving the origin of the neighborhood from pixel to pixel and applying the operator T to the pixels in the neighborhood to yield the output at that location.
- $\therefore$ for any specific location (x,y) the value of the output image at those coordinates is equal to the result of applying T to the neighborhood with origin at (x,y) in f .

eg $\rightarrow$ Suppose a neighborhood is of size 3×3 and operator T is defined as "Compute the average intensity of the neighborhood"

$\rightarrow$ let an arbitrary location $(100, 150)$.

$\rightarrow$ Assume origin is at centre, the result $g(100, 150)$ is the sum of $f(100, 150)$ and its 8 neighbors

$\rightarrow$ i.e. divide by 9.

$\rightarrow$ The origin is then moved to the next location and the process is repeated until it covers the whole image.

Note:- The process starts at the top left of input image and proceed pixel by pixel in horizontal scan one row at a time. (15)

★ Refer ^{all} the topic of pdf 6 (PPT 1 to 24)

Numericals

Q Obtain the digital Negative of the following 3×3 image.

a)

$$f(x,y) = \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 0 & 1 & 1 \\ \hline 2 & 2 & 3 \\ \hline \end{array}$$

Since the image has 4 gray levels.

$$L = 4$$

% for digital Negative

$$S = L - 1 - r$$

$$= 4 - 1 - r = 3 - r$$

%

$$g(x,y) =$$

$$\begin{array}{|c|c|c|} \hline 2 & 1 & 0 \\ \hline 3 & 2 & 2 \\ \hline 1 & 1 & 0 \\ \hline \end{array}$$

b)

$$f(x,y) = \begin{array}{|c|c|c|} \hline 122 & 150 & 200 \\ \hline 225 & 225 & 225 \\ \hline 250 & 250 & 240 \\ \hline \end{array}$$

No. of gray level $2^8 = 256$

$$S = L - 1 - r = 255 - r$$

$$g(x,y) =$$

$$\begin{array}{|c|c|c|} \hline 133 & 105 & 55 \\ \hline 30 & 30 & 30 \\ \hline 5 & 5 & 15 \\ \hline \end{array}$$

Q Given Image. Assume the gray-level is 0-7, that is 8, apply the following transformations: Inversion, log, power law. for $c=1$ &

$$f(x, y) =$$

1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

$$v=1.2$$

(a) Inversion / Digital Negative :

$$s = L - 1 - r$$

$$\% L = 8$$

$$g(x, y) =$$

6	5	4	3
2	2	1	1
1	0	1	1
1	0	5	4

b) log

$$s = c \log(1 + r)$$

$$\% c = 1$$

$$g(x, y) =$$

0.3	0.47	0.60	

=

0	0	1	

c) Power Law

$$s = c r^v$$

$$\% c = 1, v = 1.2$$

$$g(x, y) =$$

1	2.29	3.73	5.27
6.89	6.89	8.58	8.58
8.58	10.3	8.58	8.58
8.58	10.3	2.89	3.73

=

1	2	4	5
7	7	9	9
9	10	9	9
9	10	3	4



Watermarkly



Scanned with OKEN Scanner

∴ Range is 0-7

∴

$g(x, y) =$

1	2	4	5
7	7	7	7
7	7	7	7
7	7	3	4

★ See the topic of PDF 6 (25 to

Piece wise Linear Transformation

- Since manipulation of contrast is difficult process, piecewise linear function are used to manipulate the contrast of images.
- This transformation attempt to improve image quality by manipulating the range of intensity values.
- The stretch is done to the maximum possible values to achieve best visualization of an image.
- Advantage -
 - Use simple function to represent complex function for intensity transformation.
- Disadvantage -
 - Require many input parameters from the user.

★ Contrast Stretching
from PDF 6

Range Normalization :-

- It is a special form of contrast stretching which is used to enhance image quality by altering the range of gray levels present.

- Assume gray level of image range from a to b .

If contrast is required to be changed as c to d .

then change can be accomplished by following steps.

Step 1. Subtract a from each gray level, so range now becomes
 0 to $(b-a)$

Step 2. Similarly the contrast image range
 0 to $(d-c)$

Step 3. The transformation is as follows :

$$g(x,y) = \frac{d-c}{b-a} [f(x,y) - a] + c$$

∴ The input range (a to b) can be transformed to output image range (c to d)

Q A gray level image is given, range from 10-60. It is necessary to transform this image to another image B , whose range is 120-180. What would be the gray level transformation

10	15
20	50

Soln:- $a = 10$, $b = 60$, $c = 120$, $d = 180$

(17)

$$g(x, y) = \frac{d - c}{b - a} [f(x, y) - a] + c$$

$$= \frac{180 - 120}{60 - 10} [f(x, y) - 10] + 120$$

$$= 1.2 [f(x, y) - 10] + 120$$

∴ Gray level transformation results in

$$g(x, y) = \begin{array}{|c|c|} \hline 120 & 126 \\ \hline 132 & 168 \\ \hline \end{array}$$

Gray level slicing } PDF (6)
Bit - Plane slicing }

Q Perform gray-level slicing on the following image

3	4	5
6	6	7
1	2	2

Soln:- Gray-level slicing is done as:

(a) Without background

$$s = \begin{cases} L - 1 & ; r_1 \leq r \leq r_2 \\ 0 & ; \text{otherwise} \end{cases}$$

$\therefore L = 8, \quad r_1 = 3, \quad r_2 = 6$

$\therefore$ Resulting Image

7	7	7
7	7	0
0	0	0

b) With Background

$$S = \begin{cases} L-1 & ; \quad r_1 \leq r \leq r_2 \\ r & ; \quad \text{otherwise} \end{cases}$$

$\therefore$ Resulting Image

7	7	7
7	7	7
1	2	2

eg Show the Bit-Plane slicing of the image

7	6	5
4	3	2
1	1	0

Soln:- The binary Equivalent of the pixels

110	110	101
100	011	010
001	001	000

421

- Suppose the least significant bit (LSB) is changed to 0, (18) then the image is reduced to the following form.

110	110	100
100	010	010
000	000	000

- The image can be further reduced to the following form

6	6	4
4	2	2
0	0	0

- Suppose the most significant bit (MSB) is changed to 0, then

011	010	001
000	011	010
001	001	000

- The image can be further reduced to the following form

3	2	1
0	3	2
1	1	0



Q. Apply bit-plane slicing on the following image size.

167	133	111
144	140	135
159	154	148

Step 1. The binary equivalent for above image is

10100111	10000101	01101111
10010000	10001100	10000111
10011111	10011010	10010100

2	167
2	83
2	41
2	20
2	10
2	5
2	2
2	1

For 1's digit : Binary formed of 167

1	0	1	0	0	1	1	1
MSB	7 th bit	6 th bit	5 th bit	4 th bit	3 th bit	2 nd bit	LSB
7	6	5	4	3	2	1	0

Step 2. Bit-Plane Slicing of above example

1	1	0
1	1	1
1	1	1

7th Bit
MSB Bit Plane

0	0	1
0	0	0
1	1	1

6th Bit

1	0	1
0	0	0
0	0	0

5th Bit

0	0	0
1	0	0
1	1	1

4th Bit

0	0	1
0	1	0
1	1	0

5th Bit

1	1	1
0	1	1
1	0	1

2nd Bit

1	0	1
0	0	1
1	1	0

1st Bit

1	1	1
0	0	1
1	0	0

3rd Bit

LSB Bit Plane

- Further generate a binary image by apply thresholding at 145 and write the image.

Binary image after threshold

1	0	0
0	0	0
1	1	0